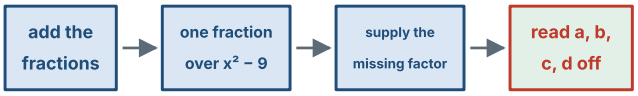
Fraction Algebra 2 — Solving Coefficients

Worked method · diagrams for every step · Desmos lines · three new questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 What the question is really asking

the two sides are the same fraction, so the numerators must match once the denominators do

Nothing here is solved for *x*. The equation holds for **every** x, so it is an identity: add the right-hand side into a single fraction, make its denominator the one you were given, and the numerator that appears **is** $ax^3 + bx^2 + cx + d$.



match the denominator you are given before you read any coefficient

The only thinking step is the third one. $(x^2 - 9)(x + 2)$ is $(x + 3)(x - 3)(x + 2)$, and adding the two fractions only gets you as far as $(x + 3)(x - 3)$ — so the missing factor is $(x + 2)$.

2 Reading the coefficients off, instead of expanding

P(0) = d | P(1) = a + b + c + d | so a + b + c = P(1) − P(0)

Once you know the numerator is some cubic P(x), you never have to expand it. Substituting x = 1 adds every coefficient up; substituting x = 0 leaves only d. **Subtract, and the d cancels.** This is the whole reason the question asks for a + b + c and not for b alone.

P(x) is the numerator, so substituting reads its coefficients off

P(0)	= d, the constant term on its own
P(1)	= a + b + c + d, all four at once
P(1) − P(0)	= a + b + c, which is what is asked

Each substitution is one line in Desmos. Expanding the cubic by hand is three lines of algebra and three chances to drop a sign.

3 The one thing that can go wrong

the shortcut needs x = 0 and x = 1 to be ordinary values, not poles

You are substituting into the **fraction** form, which is undefined wherever a denominator is zero. In this question the poles are x = 3 and x = −3, so 0 and 1 are both fine. If a question put a pole at x = 1, Desmos would return *undefined* and you would have to expand the numerator first, or fit it instead.

before you substitute, check the value is not a pole

x = 0	not a pole — safe	✓ use it
x = 1	not a pole — safe	✓ use it
x = 3 or x = −3	a pole — P is undefined there	× avoid

Check the poles before you substitute, not after Desmos returns undefined.

4 The worked question

$(ax^3 + bx^2 + cx + d) \div ((x^2 - 9)(x + 2)) = 5/(x + 3) + 6x/(x - 3)$ — find $a + b + c$

Solution. Add the right-hand side, then supply the factor the denominator is still missing.

Working.

$5(x - 3) + 6x(x + 3) = 5x - 15 + 6x^2 + 18x =$ **$6x^2 + 23x - 15$**, all over $x^2 - 9$.

$(x^2 - 9)(x + 2)$ needs one more factor of $(x + 2)$, so multiply top and bottom by it.

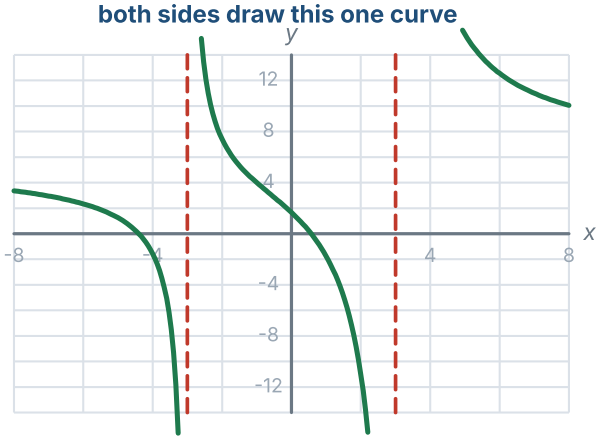
$(6x^2 + 23x - 15)(x + 2) =$ **$6x^3 + 35x^2 + 31x - 30$**.

$a = 6$, $b = 35$, $c = 31$ and $d = -30$, so **$a + b + c = 72$**.

5/(x+3) + 6x/(x−3) → 6x² + 23x − 15 → × (x + 2) → 6x³ + 35x² + 31x − 30

over $x^2 - 9$ first, then the missing $(x + 2)$: a = 6, b = 35, c = 31

both sides draw this one curve



The check worth doing: graph the cubic over the given denominator and the original right-hand side together. **One** curve means the coefficients are right; two means they are not. The dashed red lines are the asymptotes at $x = -3$ and $x = 3$, the two values the identity says nothing about.

In Desmos (five lines, and two of them are only the check)

Two lines solve it. The first names the numerator; the second reads the answer out of it. The third and fourth are the check — define the cubic you found and see that it draws the same curve.

What you are doing	Type into Desmos	Why	It shows
Name the numerator as a function of x.	`P(x)=(5/(x+3)+6x/(x-3))(x^2-9)(x+2)`	Everything to the right of the equals sign, times the denominator you were given.	—

What you are doing	Type into Desmos	Why	It shows
Read off the answer.	`P(1)-P(0)`	P(1) is a + b + c + d and P(0) is d, so the difference is a + b + c.	**72**
Check the constant term on its own.	`P(0)`	This is d. It is not part of the answer, but it tells you the expansion is right.	**−30**
Get every coefficient, not just the sum.	`y_1~ax_1^3+bx_1^2+cx_1+d`	Put x_1 = 0, 1, 2, 4, 5 in a table with y_1 = P(0), P(1), P(2), P(4), P(5), then type the regression under it. Desmos reports all four coefficients exactly, with RMSE 0. This route also works when a pole blocks the shortcut.	**a = 6, b = 35, c = 31, d = −30**
Confirm it by eye.	`y=(6x^3+35x^2+31x-30)/((x^2-9)(x+2))` and `y=5/(x+3)+6x/(x-3)`	Two expressions, one curve. If the coefficients were wrong the two graphs would separate.	**the curves coincide**

Three more of the same kind (hardest last)

Same shape every time: a cubic over a disguised product of three linear factors, equal to two fractions whose denominators are two of those three. Find the factor the sum is **missing**, and the numerator falls out.

#	Level	Question	Answer	Solution
Q1	Hard	$(ax^3 + bx^2 + cx + d) \div ((x^2 - 16)(x + 1)) = 3/(x + 4) + 2x/(x - 4)$. What is the value of a + b + c? $3/(x+4) + 2x/(x-4)$ → $2x^2 + 11x - 12$ → $\times (x + 1)$ → $2x^3 + 13x^2 - x - 12$ poles at x = 4 and x = −4, so P(1) − P(0) is safe	**14**	Add the right-hand side over $x^2 - 16$: $3(x - 4) + 2x(x + 4) = \mathbf{2x^2 + 11x - 12}$. The given denominator carries an extra (x + 1), so the numerator is $(2x^2 + 11x - 12)(x + 1) = \mathbf{2x^3 + 13x^2 - x - 12}$. a + b + c = 2 + 13 − 1 = **14**. In Desmos: P(1) − P(0) = 2 − (−12) = 14.
Q2	Hard	$(ax^3 + bx^2 + cx + d) \div ((x^2 - 25)(x - 2)) = 4/(x - 5) + 7x/(x + 5)$. What is the value of a + b + c? $4/(x-5) + 7x/(x+5)$ → $7x^2 - 31x + 20$ → $\times (x - 2)$ → $7x^3 - 45x^2 + 82x - 40$ the 4 pairs with (x + 5) and the 7x with (x − 5), not the other way round	**44**	The fractions are written the other way round, so take care which one is multiplied by which: $4(x + 5) + 7x(x - 5) = \mathbf{7x^2 - 31x + 20}$. The missing factor is (x − 2), giving $(7x^2 - 31x + 20)(x - 2) = \mathbf{7x^3 - 45x^2 + 82x - 40}$. a + b + c = 7 − 45 + 82 = **44**. In Desmos: P(1) − P(0) = 4 − (−40) = 44.
Q3	Very hard	$(ax^3 + bx^2 + cx + d) \div ((x^2 - 1)(x - 4)) = 7/(x + 1) + 2x/(x - 1)$. What is the value of a + b + c? $7/(x+1) + 2x/(x-1)$ → $2x^2 + 9x - 7$ → $\times (x - 4)$ → $2x^3 + x^2 - 43x + 28$ x = 1 is a pole, so P(1) is undefined — fit the cubic instead	**−40**	Over $x^2 - 1$: $7(x - 1) + 2x(x + 1) = \mathbf{2x^2 + 9x - 7}$, and the missing factor is (x − 4). $(2x^2 + 9x - 7)(x - 4) = \mathbf{2x^3 + x^2 - 43x + 28}$, so a + b + c = 2 + 1 − 43 = **−40**. **The shortcut breaks here.** The poles are x = 1 and x = −1, so P(1) is undefined in Desmos. Either expand as above, or fit the cubic through five safe points such as x = 0, 2, 3, 5, 6.

The one habit to build. Factorise the denominator you are *given* before you touch the fractions on the right. The answer to every question of this kind is the factor that is in one list and not the other.

Every numerator here was expanded in exact arithmetic and checked against the original identity at 400 random values of x; a + b + c was then confirmed a second way as P(1) − P(0).